Al-Balqa Applied University

جامعة البلقاء التطبيقية

Note: ***Write all the required equations (without simplification) to solve the following problems, then find the final answer.***

V-2A

Prob. 1: ***Consider the circuit shown in Fig. 1,***

1. The open circuit voltage between *a-b* terminals (V_o), when the load impedance is disconnected, is:

$(8-j4)\,//\,(5+10j)$ $\quad I_5 * 5\Omega = \boxed{V_o = 23,4-29,2j}$

2. The short circuit current (I_{sc}), when *a-b* terminals are shorted is:

$I_{sc} = \frac{V_L}{Z_L} \rightarrow \frac{96+48j}{10j} = \boxed{4,8-9,6j}$

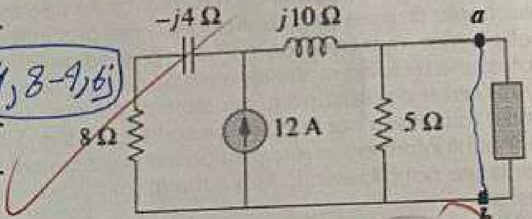

Fig.1

6

3. If the load impedance $Z_L = 8+j6\,\Omega$, The load current is:

$V_L = V_5 = 20-20j$

2

Prob. 2: Consider the circuit shown in Fig. 2 and using mesh analysis method,

1. Write the two mesh equations, do not simplify:

$I_1 \Rightarrow -40\angle 30 + 10j I_1 - 20j(I_1 - I_2) = 0$

$I_2 \Rightarrow 40 I_2 + 50 - 20j(I_2 - I_1) = 0$

2. The current I_1 is: $-0,428+4,678j$

3. The current I_2 is: $0,785+0,607j$

4. The capacitor current, I_C, is: $81,42+24,26j$

10

2

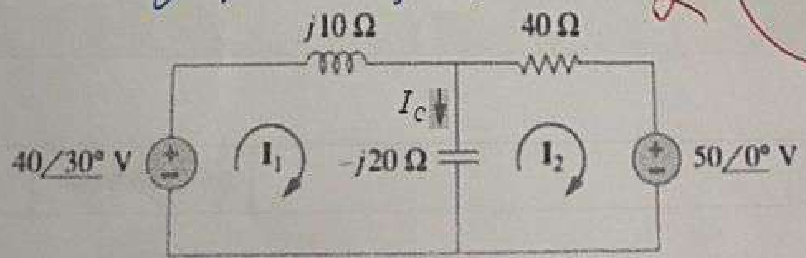

Fig 2

Consider the circuit shown in Fig. 3 .

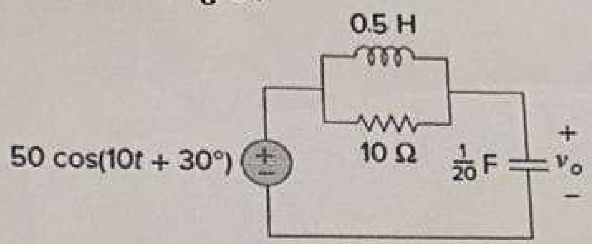

Figure 3

1. The equivalent admittance across the terminals of the source is:

$Y = \frac{1}{Z} \rightarrow 0{,}25 - 0{,}25j$

2.The voltage (V0) is:

$V_0 = I * Z_C$ ~~$V_0 = Z$~~ $\rightarrow 9{,}15 - 34{,}15j$

3.The current through 10Ω resistor in phasor domain is:

$I_{10} = \frac{V_{10}}{Z_{10}} \rightarrow 5{,}24 + 5{,}91j$

4.The total current lags the voltage of the voltage source by : 45° degrees

9